Context-Conditioned Log Writing

Step 9 — Task 3: incremental, RAG-style journal summarization

Tomer Morad Or Israeli Yoel Marko

July 2026

1. Objective

Task 3 (previously entirely unstarted): given the existing journal entry for a legislative matter and a new protocol segment about that same matter, generate *only* the incremental contribution — new developments, decisions, status changes — not a full re-summary. Faithfulness is defined as chain-relative: judged against the summary chain built so far, not by re-deriving ground truth from every prior session's raw transcript.

2. Method

Deliberately built on Step 7's gold chains, not method (a)'s predicted linking output: method (a) is only 7.7% precision at best-F1, so using its predicted chains here would conflate "is the summarizer bad" with "is the upstream linking bad." Same isolation precedent as Step 7 separating Task 1 from Task 2.

For each of the 22 gold-recurring chains (36 total incremental-update instances): generate an opening entry from the first occurrence alone, then for each subsequent occurrence, generate an update given (a) the running journal built from the model's *own* prior generations and (b) the new occurrence's text (capped at 700 words). Two generators compared: `dictalm2.0-instruct` (Hebrew-native) and `Qwen2.5-7B-Instruct`. Every entry is then scored by an independent judge (`Qwen2.5-3B-Instruct`, not used for generation, to avoid self-preference bias) on two 0–100 axes: **faithfulness** (grounded in the source text, no hallucination) and **novelty** (updates only — genuinely new content vs. just restating the existing journal).

3. Results

Prompt version	dictalm2 F.	dictalm2 N.	qwen7b F.	qwen7b N.
original (no few-shot)	**80.7**	**18.1**	63.4	**10.1**
v2 (real-corpus example)	71.1	15.0	63.1	8.3
v3 (fictional example)	53.6	5.0	**69.2**	5.3

F. = mean faithfulness, N. = mean novelty (updates only), both 0–100.

All three configurations had 0/58 JSON parse errors. **Faithfulness**: dictalm2 (Hebrew- native) is consistently more grounded than qwen7b on the original prompt — the opposite pattern from Step 8, where dictalm2 struggled with rigid JSON-schema output but this is free-text generation, playing to its strength.

Novelty is the central open problem, and two attempted fixes made it *worse*, not better. v2 added a worked BAD-vs-GOOD example built from real corpus content: 13/36 (36%) of dictalm2's updates turned out to be *literal verbatim copies* of the worked example's answer, regardless of the actual input. v3 rebuilt the example with fully fictional content plus an explicit anti-copying instruction, hypothesizing real-corpus content was the specific problem — this made copying *worse* (33/36 = 92%), traced to an authoring bug (a literal placeholder left in the fictional example's source-text slot, giving the model nothing to reason from except the fixed answer). **More importantly, independent of that bug**: novelty declined monotonically for *both* models across all three versions, including qwen7b, which had zero copying issue at all (10.1

→ 8.3 → 5.3). Few-shot examples for this specific free-text incremental-diff task cause imitation of the example's *content*, not just its *style* — the opposite of Step 8's classification task, where few-shot genuinely helped.

4. What Needs Improvement

1. **Recommendation: use the original (zero-few-shot) prompt.** It is best or tied-best on every metric for dictalm2, and few-shot didn't meaningfully help qwen7b either.
2. **Novelty scores are low across the board** (best case 18.1/100) — even without the few-shot regression, genuinely incremental (not recap-laden) log writing is unsolved.
3. **Untried next lever**: force a structured "what changed" bullet-list output instead of free prose, which should be harder to blend a copied/recapped example into than a flowing paragraph. Not attempted in this step.
4. **No human-written reference summaries exist** for this corpus, so faithfulness/ novelty are judged by an LLM against the same context the generator saw, not by comparison to a gold summary. A small human-annotated reference set (even 5–10 chains) would substantially strengthen this evaluation.
5. **Built on gold chains, not predicted linking** — by design, so this step's numbers say nothing about end-to-end pipeline quality (linking + log writing together). That combined evaluation has not been run.

Artifacts (this step):
`outputs/journal_chains.json`, `outputs/generated_logs_*.json` (6 configs),
`outputs/faithfulness_scores_*.json`, `outputs/faithfulness_summary.json`
Code: `code/build_journal_chains.py`, `code/generate_logs.py`, `code/eval_faithfulness.py`.